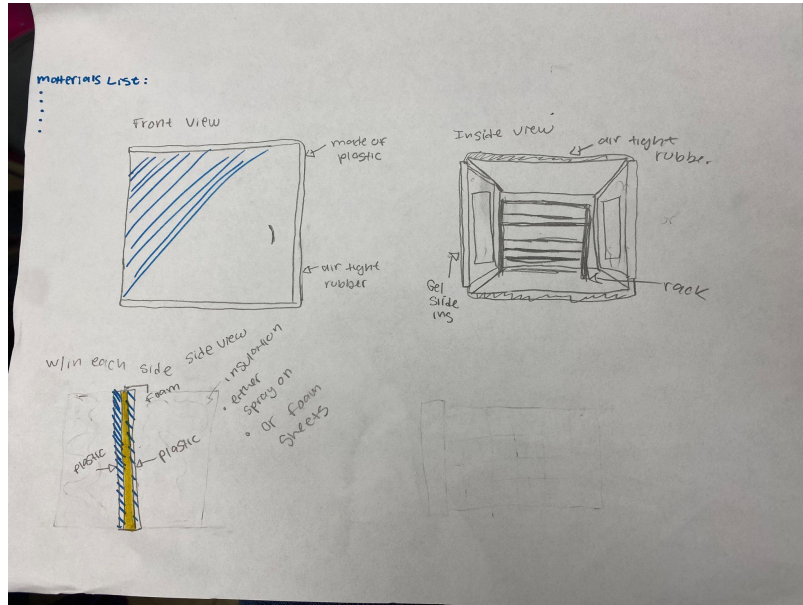


Element C

Element C

Sketch of our original solution:



Boxes can be made and be inserted into the cabinets of the art room. The walls of the boxes will have foam in the middle and plastic around the foam to make it humidity resistant. This will keep the box airtight and prevent the humidity from reaching the contents inside.

Measurements:

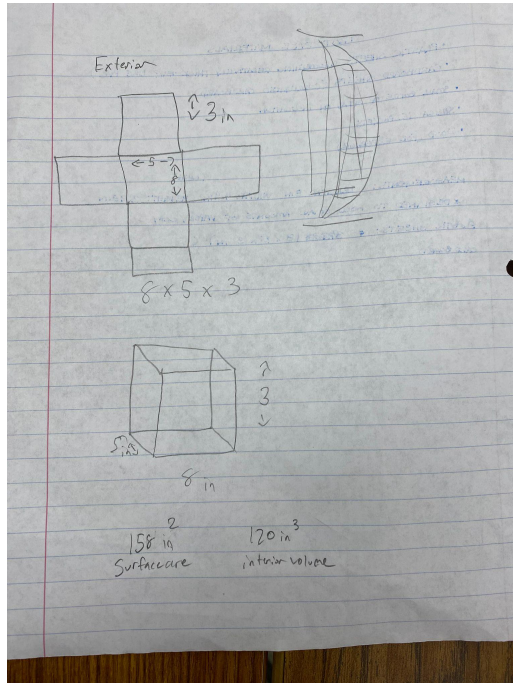
Small cabinets 23" x 30"

Per quadrant/per cubby measurement 20" x 39.4"

Other cabinet measurements 18" x 10.5"

Element C

Picture of prototype design:



Measurements for prototype (small scale $\frac{1}{3}$ to $\frac{1}{4}$ of original):

Dimensions: 8 in x 4.9 in x 2.98 in

There will be 2 types of materials for the production of the prototype. The first will be made out of plastic and the second will be made out of wood. The dimensions will be the same for each. The purpose of the small scale design is to prevent the waste of materials, ensure the design idea works before creating the final solution, and make sure the consumer is satisfied.

Essential Criteria

The student engineers decided to construct a small scale prototype to test out the suggested solution on a smaller scale before creating the final product and presenting it to the art teacher. The small scale prototype will be used to test the design without wasting materials, make sure the design works and so the art teacher and her students are satisfied with it.

Some important criteria includes:

- Keep paper and clay dry at all times (even when the humidity is strong)
- Sturdy/strong enough to hold multiple sheets of paper and clay sculptures
- Make paint from art pieces dry faster
- Size appropriate (shouldn't take too much space in the art teacher's room)
- Easily accessible for both the teacher and the students

The student engineers are sure that the solution criteria is practicable.